Muditha Fernando

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Google Scholar Homepage

Research Interests

• Computer Vision • Self-Supervised Learning • Robustness • Multi Modal Learning

Education

University of Moratuwa, Sri lanka Jan. 2020 – June 2024
B.Sc.(Hons) Electronic and Telecommunication Engineering *CGPA: 3.94/4.20*
Dean's list: Semester 1,2,6,7,8

Highlighted Courses: Machine Vision, Advances in Machine Vision, Bio-Medical Image Processing, Fundamentals of Image Processing and Machine Vision, Pattern Recognition and Machine Intelligence, Applied Statistics, Linear Algebra, Calculus, Signals and Systems, Random Signal Processing, Data Structures and Algorithms.

Experience

Computer Vision Research Group, University of Moratuwa, Sri Lanka Aug 2024 – Present
Research Student

- Self-supervised video-representation learning
- Language guided semi-supervised domain generalization (ICPR 2026)

Pekoe Pte Ltd, Singapore (Remote) Jun 2024 – Present
AI Engineer

- Researched the use of non-contrastive self-supervised learning on tea image datasets.
- Investigated supervised contrastive learning approaches for robust representation learning.
- Research on statistical feature extraction methods and texture extraction for tea images.
- Studied the applicability of masked image modeling methods to tea image datasets.
- Research on the transferability of foundational models for tea image datasets.

Pekoe Pte Ltd, Singapore (Remote) Sept 2023 – Mar 2024
Part-time AI Engineering Intern

- Conducted research on image-based methods for assessing tea sample homogeneity.
- Investigated multi-task learning approaches for feature extraction from tea images.

Pekoe Pte Ltd, Singapore (Remote) Jan 2023 – Jul 2023
AI Engineering Intern

- Applied self-supervised learning techniques for representation learning using CNN and ViT architectures.
- Investigated the use of Generative Adversarial Networks (GANs) for synthetic data generation.
- Developed time series forecasting models using SARIMAX and XGBoost.

Publications and Thesis

- **M. Fernando**, K. Kailainathan, K. Nagaratnam, I. U. B. Senavirathne, & R. Rodrigo, "SemAlign: Language Guided Semi-supervised Domain Generalization". In Proceedings of the International Conference on Pattern Recognition (ICPR 2026).
- **M. Fernando**, T. Rathnasekara, S. Nazar, A. Perera, & T. Kaluarachchi, "MaLDAM: Masked Localized Domain Adaptation for Malaria Detection in Low-Cost Microscopic Images". In Proceedings of the MICCAI 2026 Workshop: HemaRAI - Toward Reliable AI in Hematology: Robust Red Blood Cell and Parasitemia Analysis.
- **M. Fernando***, A. Perera*, T. Rajaka*, B. Yasarathna*, R. Rodrigo, S. Ramasinghe,"Perception Module for World Model"(2024). Thesis Report

  Joint first authorship indicated by *.

Research Projects

Semi Supervised Domain Generalization Jan 2025 - Dec 2025

- Aligning image representations with class label embeddings for semi-supervised domain generalization, and introduced associated loss constraints.
- Introduced a computationally efficient Fourier amplitude swapping-based data augmentation method.
- Achieves SOTA performance across multiple datasets
- Outcome: Paper, Code

Weakly Supervised Domain Adaptation for Low Cost Microscopes Mar 2026 - Jun 2026

- Cell-aware domain adaptation framework that jointly aligns global image context and masked local cellular representations via a non-contrastive objective.
- Achieved SOTA performance on the multi-microscope M5-dataset, improving mAP by up to 8.5% across magnifications without any additional inference-time latency.
- Outcome: Paper, Code

Perception Module for World Models Aug 2023 – Apr 2024

Undergraduate Thesis Project - University of Moratuwa

- Designed a shared encoder model to jointly learn content, flow, and depth features from unlabeled monocular videos using a self supervised multi task framework.
- Implemented task-specific decoders to capture spatial and temporal dependencies and improve multi-task feature consistency.

Skills

Languages: English (Professional), Sinhala (Native)
Programming Languages: Python, MATLAB, C++, System Verilog
Developer Tools: Git, VS Code, Confluence, JIRA
Libraries: PyTorch, Pandas, NumPy, Matplotlib, OpenCV, Scikit learn, ITK
Softwares: SolidWorks, Proteus, Altium, Multisim, Webots, Coppeliasim

Competitions

IEEEXtreme 15.0: 24-hour Programming Competition 2021

Secured 24th place country wide and 332nd global

E-Yantra robotics competition IIT Bombay 2021

Secured 4th place in berryminator category

Service

Teaching : Lab instructor for EN 1094 Laboratory Practical course
Mentoring: Mentor for computer vision competition teams at University of Moratuwa, Sri Lanka

References

Dr. Ranga Rodrigo
B.Sc. Eng. Hons. (Moratuwa), M.E.Sc. (Western),
Ph.D. (Western), MIET, MIEEE

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